FLATNESS AT A POINT OF A MORPHISM OF ALGEBRAIC STACKS IS CHARACTERISED BY LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The characterization of flatness at a point of a morphism of algebraic spaces through local rings. Invoked inside the counted proof (source lines 5536-5545) as a cross-chapter pointer. Supplies the local ring criterion in the scheme-free setting.
- The description of the local rings of an algebraic stack through a smooth presentation. Used inside the counted proof (source lines 5527-5535). Compares the local rings of the stack with those of the presentation.
- The statement that flatness is local on the source and target in the smooth topology. Invoked at the end of the counted proof (source lines 5546-5557). Allows the property to be moved between the stack and its presentation.

Source reference. Original lemma lemma-base-change-flat: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5376–5376. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-flat, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-base-change-module-flat: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5428–5428. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-base-change-module-flat, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference spaces-morphisms-lemma-composition-module-flat: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5550–5550. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-composition-module-flat, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-flat-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5451–5451. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-flat-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-points-cartesian: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2126–2126. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-points-cartesian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $x \in |\mathcal{X}|$. Consider commutative diagrams*

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array} \quad \text{with points} \quad \begin{array}{c} u \in |U| \\ \downarrow \\ x \in |\mathcal{X}| \end{array}$$

where U and V are algebraic spaces, b is flat, and $(a, h) : U \to \mathcal{X} \times_{\mathcal{Y}} V$ is flat. The following are equivalent

(1) *h is flat at u for one diagram as above,*
(2) *h is flat at u for every diagram as above.*

Proof. Suppose we are given a second diagram U', V', u', a', b', h' as in the lemma. Then we can consider

$$\begin{array}{ccccc} U & \longleftarrow & U \times_{\mathcal{X}} U' & \longrightarrow & U' \\ \downarrow & & \downarrow & & \downarrow \\ V & \longleftarrow & V \times_{\mathcal{Y}} V' & \longrightarrow & V' \end{array}$$

By Properties of Stacks, Lemma stacks-properties-lemma-points-cartesian there is a point $u'' \in |U \times_{\mathcal{X}} U'|$ mapping to u and u'. If h is flat at u, then the base change $U \times_V (V \times_{\mathcal{Y}} V') \to V \times_{\mathcal{Y}} V'$ is flat at any point over u, see Morphisms of Spaces, Lemma spaces-morphisms-lemma-base-change-module-flat. On the other hand, the morphism

$$U \times_{\mathcal{X}} U' \to U \times_{\mathcal{X}} (\mathcal{X} \times_{\mathcal{Y}} V') = U \times_{\mathcal{Y}} V' = U \times_V (V \times_{\mathcal{Y}} V')$$

is flat as a base change of (a', h'), see Lemma lemma-base-change-flat. Composing and using Morphisms of Spaces, Lemma spaces-morphisms-lemma-composition-module-flat we conclude that $U \times_{\mathcal{X}} U' \to V \times_{\mathcal{Y}} V'$ is flat at u''. Then we can use composition by the flat map $V \times_{\mathcal{Y}} V' \to V'$ to conclude that $U \times_{\mathcal{X}} U' \to V'$ is flat at u''. Finally, since $U \times_{\mathcal{X}} U' \to U'$ is flat at u'' and u'' maps to u' we conclude that

$U' \to V'$ is flat at u' by Morphisms of Spaces, Lemma spaces-morphisms-lemma-flat-permanence. □